

Do All MobileNets Quantize Poorly? Gaining Insights into the Effect of Quantization on Depthwise Separable Convolutional Networks Through the Eyes of Multi-scale Distributional Dynamics

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Abstract

As the “Mobile AI” revolution continues to grow, so does the need to understand the behaviour of edge-deployed deep neural networks. In particular, MobileNets [9, 22] are the go-to family of deep convolutional neural networks (CNN) for mobile. However, they often have significant accuracy degradation under post-training quantization. While studies have introduced quantization-aware training and other methods to tackle this challenge, there is limited understanding into why MobileNets (and potentially depthwise-separable CNNs (DWSCNN) in general) quantize so poorly compared to other CNN architectures. Motivated to gain deeper insights into this phenomenon, we take a different strategy and study the multi-scale distributional dynamics of MobileNet-V1, a set of smaller DWSCNNs, and regular CNNs. Specifically, we investigate the impact of quantization on the weight and activation distributional dynamics as information propagates from layer to layer, as well as overall changes in distributional dynamics at the network level. This fine-grained analysis revealed significant dynamic range fluctuations and a “distributional mismatch” between channelwise and layerwise distributions in DWSCNNs that lead to increasing quantized degradation and distributional shift during information propagation. Furthermore, analysis of the activation quantization errors show that there is greater quantization error accumulation in DWSCNN compared to regular CNNs. The hope is that such insights can lead to innovative strategies for reducing such distributional dynamics changes and improve post-training quantization for mobile.

1. Introduction

The past decade has seen an enormous boom in deep learning research, particularly in the fields of NLP and computer vision. As a result, deep learning algorithms such as convolutional neural network (CNN) models have become

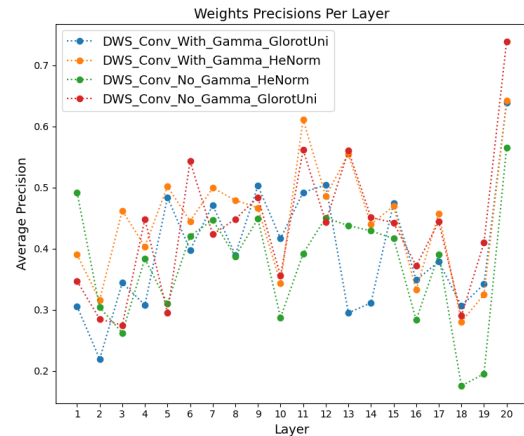


Figure 1. Layerwise average precision (see Eq. 2) of trained weights in the DWS-Convnets. Low average precision at depthwise-conv layers indicate a mismatch between each individual channel’s dynamic range and the entire tensor’s range. The resulting distributional dynamics lead to significantly higher accumulation of quantization errors. We see a similar pattern in the batchnorm-folded weights and activations of DWSCNNs.

more accessible than ever. Mobile devices have become a primary platform on which CNNs have rapidly proliferated. “AI on-the-edge,” has driven an increasing demand for deploying fast, power-efficient CNNs that can maintain highly accurate performance while operating in a resource-constrained environment. Consequently, various avenues of research have looked at making CNNs efficient enough to “fit” on mobile devices. Methods such as efficient CNN architecture design [9, 22, 8, 2, 10, 32, 29], weight pruning [5, 16, 28, 17, 15], and quantization [12, 5, 20, 21] are all aimed at reducing the storage, computation, and memory requirements of CNN algorithms. Among these methods, depthwise separable convolutional networks (DWSCNN) such as in MobileNets [9, 22, 8] and fixed point integer/quantized inference have become ubiquitous tools for designing efficient “Mobile AI” algorithms.

As Mobile AI continues to proliferate, so does the need to better understand the behaviour of the models